# APDIO Workshop: Optimization Software & Practice

## Part 2: Row Generation

João Dionísio

March 27, 2026

# Roadmap

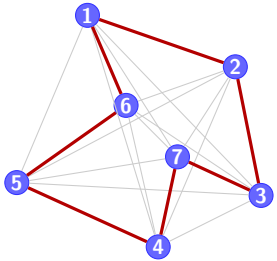
**TSP** → **Compact (MTZ)** → **Edge + Row Generation** → **Comparison**

- **Motivation:** some formulations have exponentially many constraints
- **Row generation:** add **constraints** dynamically as they are violated
- **SCIP plugin:** the constraint handler
- **Application:** Traveling Salesman Problem (TSP)

# Traveling Salesman Problem (TSP)

**Given:** $n$ cities, pairwise distances $d_{ij}$
**Find:** shortest tour visiting each city exactly once, returning to start

- One of the most famous NP-hard problems
- Symmetric TSP: $d_{ij} = d_{ji}$ (undirected edges)
- We will compare two formulations:
  1. Compact MTZ (polynomial constraints, weak LP)
  2. Extended DFJ (exponential constraints, strong LP)

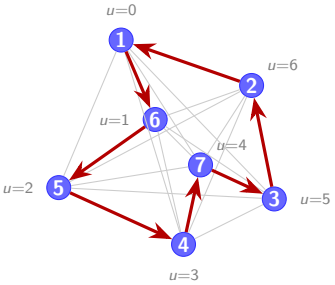



Optimal tour in red;

# MTZ Formulation

**Variables:** $x_{ij} \in \{0,1\}$ (directed edges), $u_i \in \mathbb{R}$ (positions)

$$
\begin{aligned}
\min \quad & \sum_{i \neq j} d_{ij}\, x_{ij} & & \\
\text{s.t.} \quad & \sum_{j \neq i} x_{ij} = 1 & \forall\, i \quad & \text{(leave once)} \\
& \sum_{i \neq j} x_{ij} = 1 & \forall\, j \quad & \text{(enter once)} \\
& u_i - u_j + n \cdot x_{ij} \leq n - 1 & \forall\, i, j \neq 0 \quad & \text{(MTZ)} \\
& x_{ij} \in \{0,1\} & &
\end{aligned}
$$




$u_i$ values enforce a single directed tour

- $O(n^2)$ variables and constraints
- LP relaxation is **very weak**

# Reference: Compact MTZ

**Already Implemented**

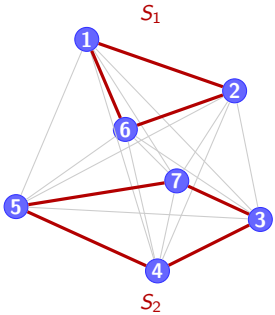
Study the MTZ formulation as a baseline before implementing row generation.

- File: `part2/compact_mtz.py`
- Run: `cd part2 && python compact_mtz.py`

# Edge Formulation with SECs

**Variables:** $x_e \in \{0,1\}$ for each undirected edge $e$

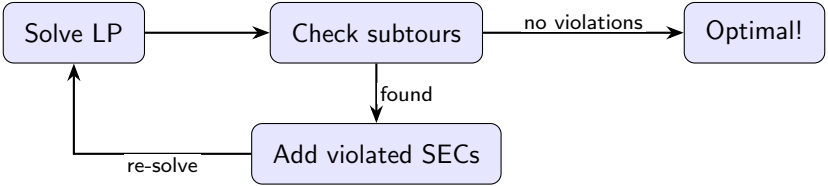
$$
\begin{aligned}
\min \quad & \sum_{e \in E} d_e \, x_e \\
\text{s.t.} \quad & \sum_{e \in \delta(v)} x_e = 2 && \forall\, v \in V \quad \text{(degree)} \\
& \sum_{e \in E(S)} x_e \leq |S| - 1 && \forall\, S \subset V \\
& 2 \leq |S| \leq n-1 && \text{(SECs)} \\
& x_e \in \{0,1\}
\end{aligned}
$$




**Subtour violation!**

- **Exponentially many** SECs: $O(2^n)$ subsets
- **Much stronger** LP relaxation than MTZ

# Row Generation Paradigm




- Start with a relaxed model (only degree constraints)
- Iteratively find and add violated constraints
- Converges to optimality without enumerating all $2^n$ SECs

# Constraint Handler in SCIP

SCIP implements row generation via the **constraint handler** plugin.

| Callback | Purpose |
|---|---|
| `conscheck` | Check if a candidate integer solution is feasible |
| `consenfolp` | Enforce constraints on LP solution; add cuts if violated |

```
class SubtourElimination(Conshdlr):
    def conscheck(self, constraints, solution, ...):
        # Find subtours; return FEASIBLE or INFEASIBLE
    def consenfolp(self, constraints, ...):
        # Add SEC for each subtour; return CONSADDED
```

# Exercises: Subtour Detection & Constraint Handler

## Exercise 1: Subtour Detection

Implement `find_subtours()`: find connected components in the edge graph.

- File: `part2/subtour.py`
- Test: `python part2/test_subtour.py`

## Exercise 2: Constraint Handler

Implement `conscheck` and `consenfolp` callbacks.

- File: `part2/conshdlr_subtour.py`
- Test: `python part2/test_tsp.py`

# MTZ vs. Row Generation

## Run the experiments

`python experiments.py` compares both formulations on increasing sizes.

**MTZ (compact)**

- $O(n^2)$ constraints
- Very weak LP bound
- Many B&B nodes needed

**Row generation (SECs)**

- Exponential constraints, added lazily
- LP bound often near-optimal
- Far fewer nodes

**Key insight:** The LP relaxation bound is what matters. MTZ's LP is significantly weaker, while the SEC LP bound is tight. This gap widens as instance size grows.

# Summary

**What we covered:**

- TSP: compact MTZ vs. edge formulation with SECs
- Row generation paradigm: add constraints on-the-fly
- SCIP constraint handler plugin (`conscheck`, `consenfolp`)
- Exponential constraint families handled implicitly

**Key takeaway:**

- A strong LP relaxation can make a substantial difference in solving time
- Row generation lets us use formulations with exponentially many constraints

**Part 3:** Branch-and-price for bin packing